FINAL ASSIGNMENT

Final project: add one useful feature

Your job: choose one change, build it, hand it over

1 · Choose

Name a coordinator problem.
Agree one small Python or SQL
change with the instructor.

2 · Build

Use the supplied baseline.
Apply the Python idea you need.
Write two expected examples.
Implement and check the change.

3 · Hand over

Let a peer run it.
Fix or explain their finding.
Provide a usable run guide.

A notebook function or SQL query can be enough; a web interface is optional.

Where to work each week

Week 8 · Propose

Open Week_08.ipynb.
Use “Project preparation for Week 9.”
Write your need and two cases.

Week 9 · Implement

Open Week_09.ipynb.
Run the supplied setup.
Write the feature under “Your turn” in Project Studio.

Week 10 · Demonstrate

Open Week_10.ipynb.
Use the operator demo in Session 2, then individual verification in Session 3.

No new Research assignment is added in Weeks 8–10.

First write this short project proposal

Answer	A concrete example
Who needs it?	The university pickup coordinator.
What is difficult?	Finding accepted pickups with no collection record.
What will you add?	A list of pickup IDs still waiting for collection evidence.
What stays outside scope?	No new events or stock changes; this is a read-only list.

This is one example. Choose your own approved small need; do not rebuild the baseline.

Before coding: write at least two input/result examples

Miniature fixture / action	Expected observation	Why this case matters
R-A accepted; no events	Include R-A	Normal pending collection
R-B accepted; collected only	Exclude R-B	Collection differs from delivery
Run report twice	Same result; state unchanged	Read-only behavior

Write independent expected outcomes, then compare the actual result.

Then implement and check your feature

1

1 · Start clean

Run the supplied fixture and baseline setup cells.

2

2 · Add your change

Write the function or query in your own project cells/module.

3

3 · Compare

Run your two examples. Compare actual and expected results.

4

4 · Protect old work

Check an unaffected case and a boundary/failure case.

For a read-only feature, also show that the database stays unchanged.

Example: what would make the collection list wrong?

Input situation	Expected list
Accepted pickup; no collection	Include its ID.
Collected but not delivered	Exclude its ID: collection has already happened.
Already delivered	Exclude its ID.
Run the report again	Same IDs; no database rows changed.

For this example feature, the unchanged supplied fixture should list P06 only.

Ask a peer to try it without your narration

1

Give

Starting data, run steps and the expected results.

2

Observe

A peer follows the steps and tries one additional input.

3

Record

Write what happened and what was confusing or wrong.

4

Respond

Fix it or explain your decision, then rerun the check.

Keep the peer finding, your response and the retest result.

Save a project folder another person can run

Include	What it must let the reader do
Code + required data	Run the approved feature.
README / run guide	Directory, reset, run steps and expected output.
Checks and results	Inputs, predictions and observed results.
Peer-review note	Finding, your response and the retest.
Limit + recovery step	One limitation and what to check after failure.
Individual note	Your contribution, one decision and one test.

Submit to submit.progmis@gmail.com and keep a backup.

At the final demonstration

Show

Explain the problem.
Run the baseline and your changed behavior from known starting data.

Let another person try

Give a peer the guide.
Let them use an unseen input.
Compare the observation with your expectation.

Explain individually

Each member traces a result
or makes a small approved change.
Explain a decision, test and revision.

The result, the run instructions and your understanding all matter.

Assessment evidence and pending details

Assessed evidence

Scope and implementation
Verification and peer revision
Handoff and individual understanding

Still pending

Numeric split within 30%
Dates, teams and presentation slots
Private route and exact file name

Before submission

Check the final announcement.
Keep code, tests and evidence.
Retain a separate backup.

A small verified improvement can earn full standard credit; the rubric split is not yet confirmed.